The Delta Tree

A replicated list with relative ranges, isometric deletion, and an identity-arbitrated merge: the design, the refutation of its strictly dead-free merge, and the corrected design (v3)

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Abstract

This note gives the complete design of the *delta tree*: a replicated list whose state is a tree of live nodes carrying *parent-relative* fractional ranges (deltas), whose delete is an *isometric fold* (survivor positions arithmetically unchanged), and whose three-way merge is OR-set survival plus a *local* overlap-splitting repair. The design is the endpoint of a systematic exploration: every component is forced by a named, machine-checked countermodel to some alternative. It is the only candidate in its corner of the design space: *strictly dead-free* (no dead identifier survives anywhere in the state) while aiming at convergence and order preservation, with a merge simple enough to mechanize. We work through the operations and the merge on concrete examples, tabulate the full litmus battery (L1–L25) against the published tombstoned RGA and the proved tombstone-free RGA, and state the anomaly contract. We then report the verdict of randomized property-based testing over a general MRDT execution model (many replicas, random merges under the LCA discipline): **the strictly dead-free merge is refuted** — battery-clean, including every named countermodel that killed its eight predecessors, yet flipping co-displayed pairs at random-DAG scale (43/120 executions). We extract and minimize the countermodel (now litmus L25), identify the mechanism (*fold-verdict erasure*: a dead node's timestamp is load-bearing for repair verdicts it participated in), refute the one principled repair as well (*source-consistent folding*, killed by a second mechanism, *repair non-locality*), and draw the structural conclusion: the three properties {strictly dead-free, pairwise display stability, topology convergence} each have a machine-checked witness for every pairwise combination and eight failed attempts at the triple. Section 5 then gives the correction: the refutation is a refutation of the *merge*, not the design. Comparing the failing executions against `path-2`, the surviving immutable-key design, shows the reconciling information was in the state all along — the *birth-parent chains* — and the merge was arbitrating from the geometry instead, which its own repairs perturb. `delta-tree-v3` keeps every local operation verbatim, adds a one-pointer-per-node birth ledger, and rebuilds the merge on three invariants (arbitrate from identity only; geometry is a rendering; render/carve compatibility). It is machine-checked clean on the full battery (except one-sided L19, shared by every one-sided design) and on the randomized DAG model at 120/120 and 300/300 executions. The triangle stands — v3 pays it one immutable parent id per node, consulted only at merges — and the two surviving designs turn out to be layers of one: identity for deciding, geometry for reading.

1 The design

State. A finite map from timestamps (which double as element identities) to records

$$\sigma : t \ \mapsto \ (\,\mathrm{par},\ (lo, hi),\ \mathrm{char}\,), \qquad 0 \le lo < hi \le 1,$$

where (lo, hi) is the node's range *relative to its parent's range* — the *delta*. The root sentinel 0 owns the unit interval and is never stored. A node's *absolute* range is the affine composition of the deltas along its root path. The state contains **live nodes only**, and — the design's distinguishing ambition — **no dead identifier persists anywhere**: not as an entry, not as a name inside a field.

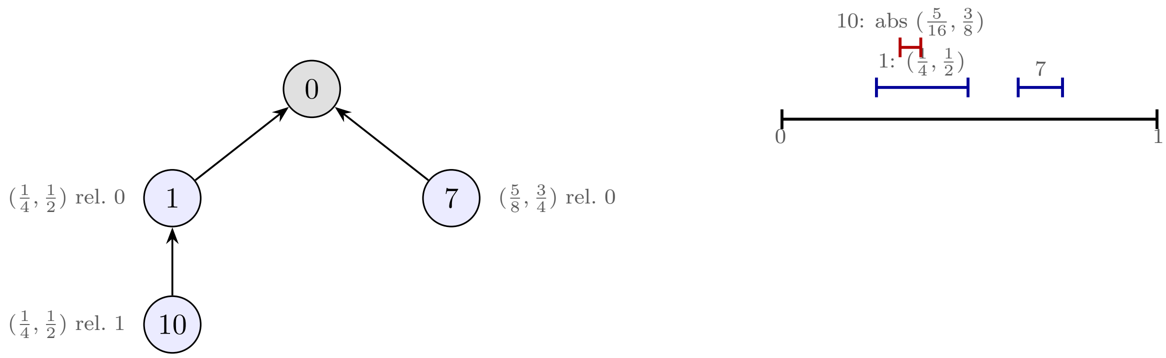


Figure 1: The state is a tree of deltas (left); absolute ranges are composed down the path (right): 10's absolute range is $\frac{1}{4} + \frac{1}{4} \cdot (\frac{1}{4}, \frac{1}{2}) = (\frac{5}{16}, \frac{3}{8})$. Display order: depth-first, children by descending range — newest first.

Insert. *Carve*: the new child of anchor a takes a slice of a's free child-space above the current head — with remaining relative gap $(b, 1)$ (where b is the current head's hi, or 0), the newcomer takes $\left(b+\frac{w}{4},\, b+\frac{w}{2}\right)$, $w = 1-b$, leaving room above for future siblings and below for its own children. Newer siblings sit higher; the display (descending) shows them first — the RGA convention, realized geometrically instead of by timestamp comparison at read time.

The operation payload (per the proved RGA). Following the *why-the-path-matters* analysis: `rc` must be `Either` everywhere (any oriented insert/delete pair awakens the refuted `cond_comm_base`), so the insert carries its anchor's ancestor *id-path* plus **one** absolute slice — the composed delta at mint time. Ids suffice for the path because the isometric fold (next) makes the absolute position invariant under ancestor deletion: a reordered application resolves the id-path to the nearest live ancestor and re-relativizes the carried absolute — landing exactly where insert-then-delete would have put it. On reachable states the anchor is live and none of this is read: the path is ghost state in the strict, machine-checked sense (`ins_path_free`).

Delete: the isometric fold. Remove the record; re-parent each child to the grandparent with the dead node's delta composed in: $c.\text{frac} := d.\text{frac} \circ c.\text{frac}$. Every survivor's absolute position is unchanged — *delete-order preservation is an arithmetic identity*, where the flat RGA's climb re-sorts (the $[b, a, c] \to [c, b]$ anomaly that started this whole exploration) and the rose tree's splice needed sixteen thousand lines it ultimately could not have.

2 The merge, strictly dead-free (as originally specified)

This is the merge as first designed — arbitrating from geometry, keeping the state strictly dead-free. It is the component §4 refutes and §5 replaces; it is retained here in full because the refutation is only meaningful against the faithful artifact. merge(L, A, B), L the least common ancestor:

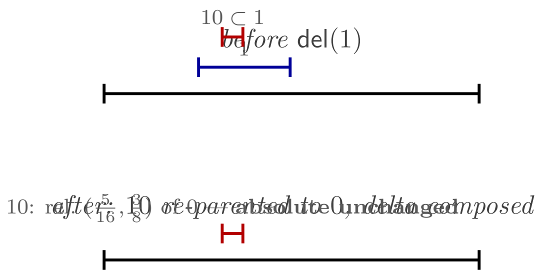


Figure 2: The isometric fold. Deleting 1 rewrites 10's delta to $(\frac{1}{4}, \frac{1}{2}) \circ (\frac{1}{4}, \frac{1}{2}) = (\frac{5}{16}, \frac{3}{8})$ relative to the root: the composition that *was* its absolute range. Nothing observable moves, and no trace of 1 remains.

1. **Survival** (OR-set): live iff in all three or born in exactly one branch.

2. **Values**: shared nodes (all in L, by the LCA discipline) take the *LCA's* (par, frac) — the canonical, topology-free base frame, deliberately reverting branch-local repairs; branch-born nodes bring their branch's. *Why revert?* Branch values encode private merge history; computing the repair over them makes the output depend on merge order — the machine-checked failure of global re-ranging (litmus L22). The LCA's values are the unique data both branches share. Because a repair is a deterministic function of the overlap family's union and its members' timestamps, reverting loses nothing: the merge *recomputes* the branch's repair, canonically extended (§4 tests this claim).

3. **Dead-chain folding**: a survivor whose parent is merge-dead folds through the dead chain using the LCA's records (dead-in-merge ⇒ live-in-LCA: the LCA does the remembering, as in the proved RGA — but isometrically, not by re-sorting).

4. **Local repair** (the overlap split): per parent, in relative coordinates, connected components of overlapping child ranges have their *union* re-split into equal slots in timestamp order, newest topmost; only the members' own fractions are rewritten — descendants are relative and follow automatically ($O(1)$ per member: the delta payoff). Singletons untouched; no global rebalance; recurse.

Worked example. Branches carve twins: L empty; A: $\mathsf{ins}(10 \leftarrow 0)$, B: $\mathsf{ins}(6 \leftarrow 0)$ — identical computations, identical slices $(\frac{1}{4}, \frac{1}{2})$. Merge: family $\{10, 6\}$, union $(\frac{1}{4}, \frac{1}{2})$, timestamp order $10 > 6$: $10 \mapsto (\frac{3}{8}, \frac{1}{2})$, $6 \mapsto (\frac{1}{4}, \frac{3}{8})$; display $[10, 6]$. The repair wrote two fractions; had 6 carried a subtree, it would have moved with it.

3 Litmus results and the anomaly contract

The suite (`whiteboard/litmus/`, tests L1–L25 with provenance) ran the faithful implementation. "strict" = the strictly dead-free merge of §2; "v3" = the corrected design of §5; "RGA$_\dagger$" = the published, tombstoned RGA; "RGA$_{tf}$" = the repo's *proved* tombstone-free RGA.

test	strict	v3	RGA$_\dagger$	RGA$_{tf}$	what it detects
L1 delete-reorder	✓	✓	✓	×	sequential splice re-sort
L2 splice fooling pair	✓	✓	✓	×	delete erases the side bit
L3a/b deep delete-runs	✓	✓	✓	✓	transitive testimony loss
L4 criss-cross split	✓	✓	✓	✓	the rose tree's refutation
L7 concurrent runs (g)	✓	✓	✓	✓	forward interleaving
L17/L18 ceiling escapes	✓	✓	✓	✓/×	eager-scalar staleness
L19 backward runs (h)	×	×	×	×	prepend interleaving (one-sided)
L20 tie inheritance	✓	✓	✓	×	nameless carving
L21/L22/L23 topology	✓	✓	✓	✓	re-ranging / frame leaks
L24 frame mixing	✓	✓	✓	✓	cross-frame siblings
L25 fold-verdict	×	✓	✓	×	**the strict merge's countermodel**
DAG PBT (§4)	× (43/120)	✓(120/120, 300/300)	✓	×	random branches and merges

Intended contract (the anomaly-matrix columns): strictly dead-free (a, *strict* — unique among live candidates), sequential soundness (c), pairwise display stability (d), forward non-interleaving (g) — with *licensed*: strong-list (e) and oracle fidelity (f), both provably impossible tombstone-free, and backward runs (h), the shared one-sided limitation. The battery certifies everything above the line. The two rows below the line are the subject of the next two sections: for the strict merge **the claimed column d fails**, in a shape no prior test exercised (§4); the corrected merge restores it at the cost of strictness (§5).

4 The execution model, and the refutation

The model. Randomized version-DAG executions (`litmus/pbt.py`): n replicas; per round each replica either issues honest operations against its own read or merges with a peer — legal only when some recorded version's event set equals the two heads' intersection (the LCA discipline); final convergence rounds drive all replicas to the full event set along different merge paths. Checks: **FLIP** (global pairwise display stability: across every read of every replica, no pair is ever shown in both orders), **CONV** (equal event sets ⇒ equal reads), **LIVE/DUP** (survival correctness). 120 executions, 4 replicas, 8 rounds.

Verdict. 43 of 120 executions flip a co-displayed pair. The battery-clean strict merge fails at random-DAG scale — in shapes involving a repair, a delete of a repaired node, and a merge with a branch that had witnessed the repair's verdict.

The minimized countermodel (now litmus L25). Five nodes.

The arithmetic of the flip: in the M_1 line, the repair left $10 = (\frac{3}{8}, \frac{1}{2})$ and $6 = (\frac{1}{4}, \frac{3}{8})$; 22's *relative* slice under 6 is $(\frac{5}{8}, \frac{3}{4})$ (it carved above 16). The fold uses the LCA's $6 = (\frac{1}{4}, \frac{1}{2})$:

$$22 \;\mapsto\; \tfrac{1}{4} + \tfrac{1}{4} \cdot (\tfrac{5}{8}, \tfrac{3}{4}) \;=\; (\tfrac{13}{32}, \tfrac{7}{16}) \;\subset\; (\tfrac{3}{8}, \tfrac{1}{2}) = 10.$$

Folded 22 lands *inside* 10's repaired slot; the repair re-splits the family $\{10, 22\}$ by timestamp — and $22 > 10$, so 22 goes on top: $[22, 10, 16]$, flipping the co-displayed $(10, 22)$.

The mechanism: fold-verdict erasure. The repair verdict "10 before 6" is a fact about the *pair of timestamps* $(10, 6)$. Strict dead-freedom erases 6 — its timestamp with it — so 6's heir re-litigates the slot with *its own* timestamp, and wins a contest its parent had lost. The dead

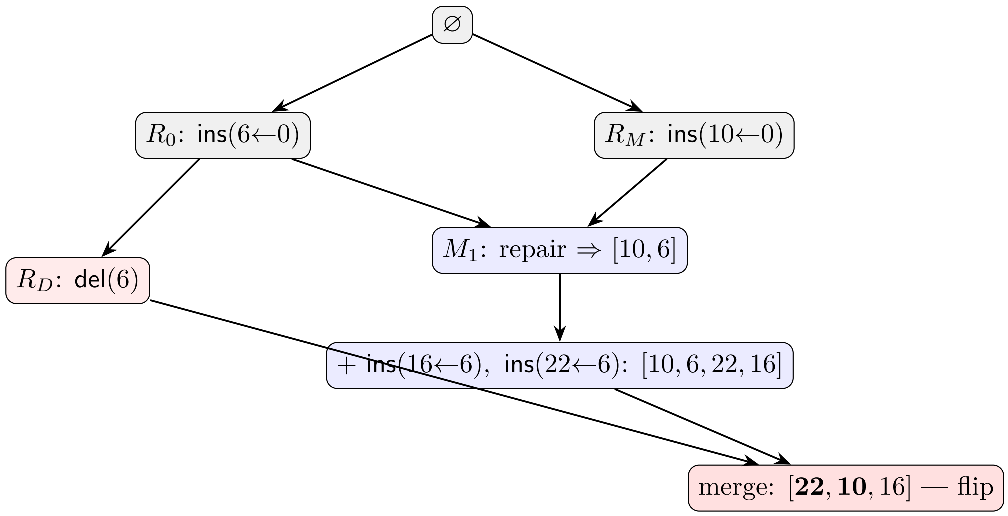


Figure 3: L25. The M_1 repair decided 10 before 6 (timestamps: $10 > 6$); typing 22 under 6 co-displayed 10 before 22. The deleter branch R_D knew only $\{6\}$. At the final merge, 6 is dead and 22 folds through *the LCA's* record of 6 — the frame that never heard the verdict.

node's timestamp was load-bearing for every order it participated in. This is the nameless-carving lemma (L20) reaching the fold: pairwise display stability requires the loser's *name* to persist, in some form, as long as the verdict matters.

The repair candidate, also refuted. *Source-consistent folding* — fold each node through the same input that supplied its value, so the fold uses the branch's (verdict-bearing) frame. It fixes L25 and the battery. The DAG PBT kills it too (41/120), by a second mechanism: **repair non-locality**. Two concurrent twins (10, 4) are tie-decided by timestamp at one merge ([10, 4], on a replica whose history is disjoint from both final inputs); elsewhere, 4 is re-slotted by repairs against *unrelated* nodes; at a causally disjoint merge the pair's geometries no longer tie, and position order — not the timestamp verdict — decides: [4, 10]. Re-slotting inside one family perturbs relations outside it; verdicts encoded in geometry are not stable under other verdicts.

5 The corrected design: identity-arbitrated merge (v3)

The refutation of §4 was answered by a conjecture (KC): *the design is almost right; the merge is not observing some invariants* — with a directive: compare `path-2`'s execution with this design's at a flipping trace, and ask whether the information needed to reconcile the order is *in the state*. It is.

The diagnosis. At a flipping PBT execution, the offending pair is first co-displayed at an insert under a node d and flips at del(d). `path-2` holds the order through the identical history because its key for the newcomer is the immutable ancestor chain *through the dead d* — position data that no repair, delete, or merge ever rewrites. The delta-tree state at the same point contains the same information: every node's birth parent is known the moment it is created, and the chain of birth parents *is* the one-sided path key. But none of the refuted merges ever consulted it. Every one of them arbitrated sibling order from *geometry* — current fractions and folded frames — and geometry is exactly what the merges' own repairs perturb (mechanism 2) and what folding through the wrong frame corrupts (mechanism 1). The correction separates the two roles the fractions had been forced to play.

The three invariants. Violated by every merge of §4; observed by v3:

1. **Arbitration from identity only.** Every order decision at a merge is computed from immutable birth-parent chains — never from current fractions, never through frame-mixed folds. The chains subsume both refutation mechanisms: a dead node's timestamp persists in its descendants' chains, so every verdict it participated in remains derivable, identically, everywhere (the L25 cure); and chains never move, so verdicts cannot be re-decided by drifted geometry at causally disjoint merges (the non-locality cure).
2. **Geometry is a rendering.** The merge re-derives all fractions, realizing the canonical order; reads remain purely geometric and never consult the chains. Between merges the geometry is authoritative for *display*; it is never authoritative for *arbitration*.
3. **Render/carve compatibility.** The re-render reproduces the sequential carve's geometry — oldest lowest, quarter slices, headroom above — so a post-merge state is indistinguishable from a sequentially carved one, and the next local carve behaves exactly as §1 specifies. (An intermediate fix violated only this invariant: its render filled each parent's space to 1, the next carve minted a zero-width slice, and a lo-tie flipped a pair — 2/120 executions. Merge output must be in the image of the local operations.)

The components. Only the state gains a field and the merge is rebuilt; **every local operation of §1 is verbatim**.

- **State.** $\sigma = (\rho, \Lambda)$: the *rendering* $\rho : t \mapsto (\text{par}, (lo, hi), \text{char})$ exactly as §1 (live nodes only), plus the *ledger* $\Lambda : t \mapsto$ birth parent — written once at insert, never mutated, retained through delete. Retention scope: an entry is needed only while some survivor's chain passes through it (live-reachable), the same stability-gated compaction as the isometric fold.
- **init**: $(\varnothing, \varnothing)$.
- $\mathsf{ins}(x \leftarrow a)$: carve into ρ as in §1; $\Lambda[x] := a$.
- $\mathsf{del}(d)$: isometric fold in ρ as in §1; Λ untouched.
- **read**: geometric DFS on ρ, descending ranges; Λ is never read.
- $\text{merge}(L, A, B)$:
    1. *Ledger union*: $\Lambda' = \Lambda_L \cup \Lambda_A \cup \Lambda_B$ — entries are immutable, so the union is conflict-free.
    2. *Survival*: OR-set, as before.
    3. *Live parent*: each survivor re-parents to its nearest surviving ancestor, found by walking Λ' — the fold's re-parenting, computed from identity instead of frames.
    4. *Canonical order*: per live parent, children sort by their **birth chains** (the root path in the immutable birth tree): compare per level, newest timestamp first; an ancestor's chain is a prefix of its descendant's and sorts first. Equivalently: the canonical order is *the RGA order of the birth tree, restricted to survivors.*
    5. *Render*: re-derive all fractions realizing the canonical order by replaying the sequential carve — children bottom-to-top, oldest first, each taking $(b + \frac{w}{4}, b + \frac{w}{2})$, $w = 1 - b$ (invariant 3). Recurse.

L25, resolved. At the final merge the survivors are $\{10, 22, 16\}$ with live parent 0. Chains: 10 is root-born; 22 and 16 were born under the dead 6, so their chains run *through* 6. Comparing 10 with 22 at the first level compares the timestamps 10 vs. 6 — the dead node's timestamp, persisting in its heirs' chains, keeps deciding, and 10 wins exactly as the M_1 repair ruled. Within 6's heirs, $22 > 16$ puts 22 first. Display: $[10, 22, 16]$ — both previously co-displayed pairs preserved. The arithmetic that produced the flip (fig. 3) is still run — but only to *render* the order the chains already settled, never to decide it.

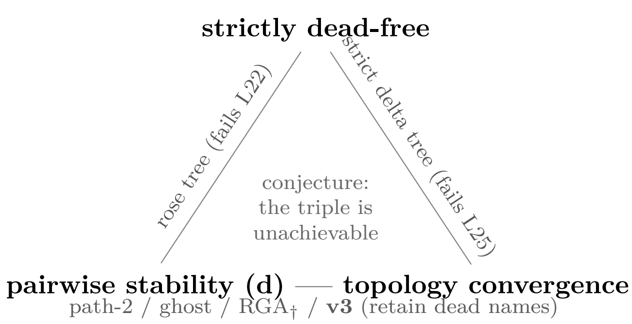
Verdict (machine-checked). `delta-tree-v3` (`litmus/delta_tree.py`) is **clean on the full battery** — including L25 and every countermodel of §3, except one-sided L19, as every one-sided design — **and clean on the randomized DAG model at 120/120** (4 replicas, 8 rounds) **and 300/300** (6 replicas, 12 rounds), plus a fresh-seed 60/60.

What v3 pays the triangle. The ledger retains dead identifiers at live-reachable scope, so v3 is not *strictly* dead-free — the triangle of §4 stands, and this is its toll. But the toll is the minimum the impossibility demands: one immutable parent id per node; no slot data, no materialized paths, no per-level ranks; nothing consulted at read time. And the two surviving designs collapse into one, layered: the birth chains are `path-2`'s one-sided key stored as a parent pointer per node instead of a materialized path; the delta tree is its runtime — $O(1)$ isometric deletes and geometric reads between merges. *Identity for deciding, geometry for reading.*

6 Conclusions

The strict merge is refuted, and the refutation is structural. Eight variants of the mutation family (scalar keys, nameless ranges, global re-ranging, local splits with and without normalization, absolute and relative storage, LCA and source-consistent folds) have now produced eight machine-checked countermodels, the last two by the two mechanisms above. The pattern admits a one-sentence summary: *order verdicts must be re-derivable, identically, at every replica and every time — and the only data that guarantees this is immutable data that names the participants, including the dead ones.*

The triangle. Every pairwise combination of the three desiderata has a machine-checked witness; the triple has eight failed attempts:



v3 does not contradict the triangle; it sits on its bottom edge with the smallest known payment — one immutable parent id per node, at live-reachable scope, never consulted by reads.

What stands. The design, with its merge rebuilt on the three invariants of §5 — and, one level down, its two permanent ideas: *(i)* parent-relative deltas are the right *representation* of a position tree — $O(1)$ subtree transport, structural sharing, lazy absolute sums — now with the immutable-position *semantics* supplied by the design's own birth chains rather than an external key scheme; *(ii)* the isometric fold is the right *compaction*: sound precisely when gated on causal stability, where "everyone has seen the delete" certifies that no verdict involving the dead node can be contested again — the condition under which both the L25 mechanism and the ledger's retention are simultaneously discharged. For mechanization, the layering suggests the proof structure directly: the chains are the specification (canonical order = RGA order of the immutable birth tree, restricted to survivors — convergence is determinism of a function of the event set), and the rendering is a refinement whose obligation is exactly invariants 2 and 3.

Provenance. Every table cell and number in this note is reproduced by `whiteboard/litmus/`: `litmus.py` (L1–L25), `pbt.py` (the execution model), `delta_tree.py` (the strict design, both fold variants, `DeltaTreeV3`, and the battery/PBT runner).